SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – SDG

Course Code:	MLA-0401	**Course Title:**	Deep Learning
CO Assessed:	CO3 – Precision (BL4)	**Assessment:**	SDG
Weightage:	30%	**Total Marks:**	100
CO3 Statement:	Implement machine learning algorithms by developing Decision Tree, Random Forest, and Support Vector Machine (SVM) classifiers using the Iris dataset, applying the Gradient Descent algorithm, and performing image segmentation using the K-Means algorithm.		
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Section / Batch:	2024	**Date:**	03/11/2026

Code of Conduct

I J. Santhosh Goud / 192425187 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *J. Santhosh Goud*

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Question 1: SDG 3 – Good Health and Well-being

Question: Design a CNN-based model for detecting pneumonia from chest X-ray images. Explain the role of convolutional layers, filters and pooling layers in the proposed system.

1. SDG Identification

Selected SDG: SDG 3 – Good Health and Well-being.

Objective: To ensure healthy lives and promote well-being for all at all ages. Within this goal, Target 3.2 aims to end preventable deaths of newborns and children under five years of age, and Target 3.8 aims to achieve universal access to quality essential health-care services. Early and accurate diagnosis of pneumonia directly supports both targets.

2. Problem Statement

Pneumonia is an infection of the lungs that fills the air sacs with fluid or pus, and it remains one of the leading infectious causes of death among children and the elderly worldwide. A chest X-ray is the most common tool for confirming pneumonia, but reading it correctly depends on the availability of trained radiologists. In rural and low-resource regions, such specialists are scarce, which results in:

- delayed diagnosis and delayed start of treatment;
- misdiagnosis caused by fatigue, workload or differences between observers, because the cloudy opacities of pneumonia can look similar to other lung conditions;
- high cost and long waiting time for reports in overcrowded hospitals.

Therefore, a fast, low-cost and reliable automated system is needed to assist doctors in screening chest X-rays for pneumonia.

3. Proposed Solution

The proposed solution is a Convolutional Neural Network (CNN) that takes a chest X-ray image as input and classifies it as **Normal** or **Pneumonia**. Unlike traditional machine learning, which needs hand-crafted features, a CNN learns the important visual patterns (lung borders, ribs, cloudy patches) directly from labelled images. The complete system works as shown in Figure 1:

- **Data:** A publicly available labelled dataset such as the Chest X-Ray Images (Pneumonia) dataset is used, divided into training, validation and test sets.

- **Pre-processing and augmentation:** Images are resized, normalised and artificially varied to make the model robust.
- **Model training:** A four-block CNN is trained using the Adam optimiser and cross-entropy loss.
- **Explainability:** Grad-CAM heat-maps highlight the lung region that influenced the decision, so that doctors can trust and verify the result.
- **Deployment:** The model is deployed in clinics or tele-health platforms as a decision-support tool; the final diagnosis is always made by the radiologist.

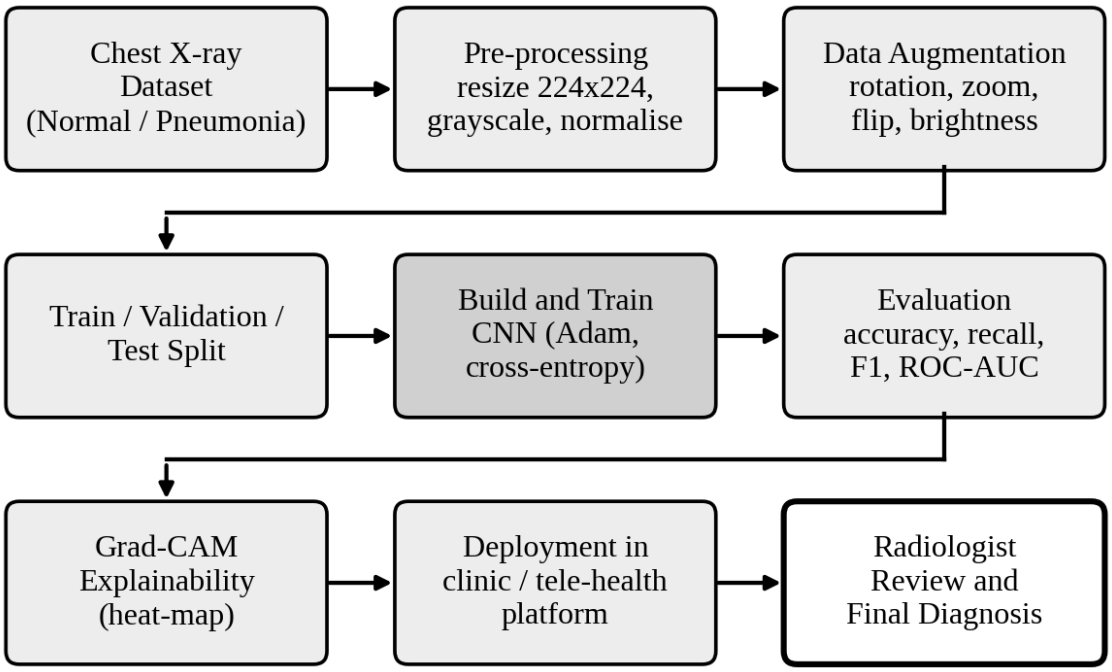


Figure 1: Workflow of the CNN-based pneumonia detection system

4. CNN Architecture

The proposed network contains four convolution + pooling blocks for feature extraction, followed by a global average pooling layer, a fully connected layer and a softmax output layer for classification. The architecture is shown in Figure 2 and the layer-wise details are given in Table 1.

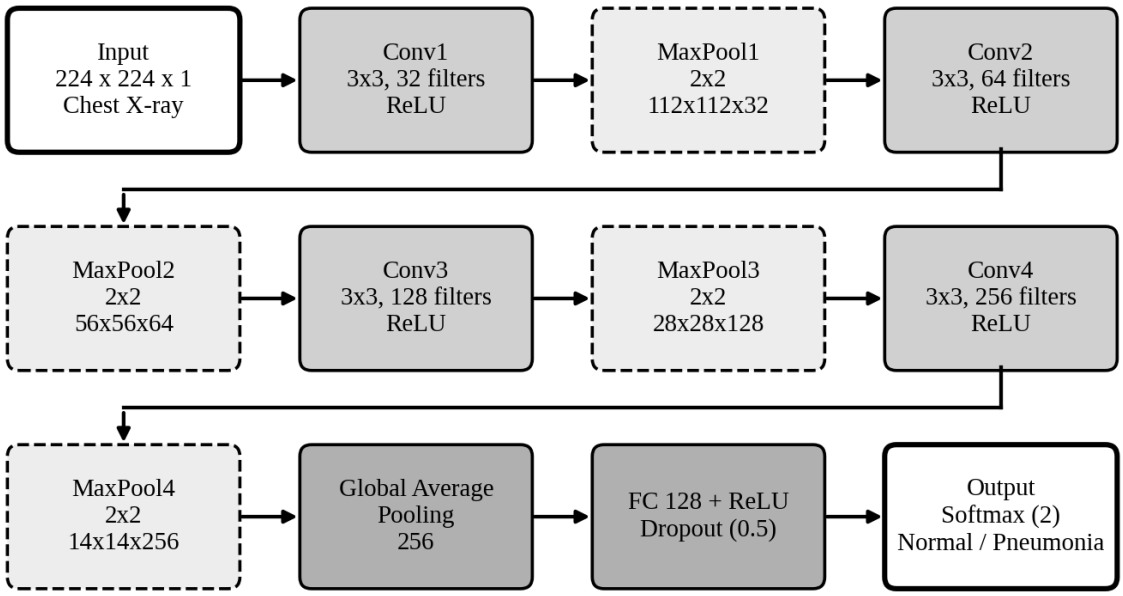


Figure 2: Proposed CNN architecture for pneumonia detection

Table 1: Layer-wise configuration of the proposed model

Layer	Configuration	Output Shape	Parameters
Input	Grayscale X-ray	224 x 224 x 1	0
Conv1 + ReLU	32 filters, 3x3, stride 1, same padding	224 x 224 x 32	320
MaxPool1	2x2, stride 2	112 x 112 x 32	0
Conv2 + ReLU	64 filters, 3x3	112 x 112 x 64	18,496
MaxPool2	2x2, stride 2	56 x 56 x 64	0
Conv3 + ReLU	128 filters, 3x3	56 x 56 x 128	73,856
MaxPool3	2x2, stride 2	28 x 28 x 128	0
Conv4 + ReLU	256 filters, 3x3	28 x 28 x 256	295,168
MaxPool4	2x2, stride 2	14 x 14 x 256	0
Global Avg. Pool	Averages each feature map	256	0
Dense + Dropout	128 neurons, ReLU, dropout 0.5	128	32,896
Output	Softmax, 2 classes	2	258

Layer	Configuration	Output Shape	Parameters
Total			**420,994**

Role of Convolutional Layers, Filters and Pooling Layers

Convolutional layers. A convolutional layer is the feature extractor of the system. A small window slides over the X-ray and, at each position, multiplies its values with the pixel values underneath and adds them up. Because the same window is reused over the whole image (parameter sharing), the network needs far fewer parameters than a fully connected network, and it can find a pattern wherever it appears in the lung. The first layers learn simple features such as edges, rib borders and the outline of the lungs. Deeper layers combine these into complex features such as hazy white patches (consolidation), fluid-filled regions and other opacities that are characteristic of pneumonia. The size of the output is given by (W – F + 2P) / S + 1; for Conv1 this is (224 – 3 + 2) / 1 + 1 = 224.

Filters. A filter (kernel) is a small matrix of learnable weights, here 3x3. Each filter detects one kind of pattern and produces one feature map; therefore 32 filters in Conv1 give 32 feature maps. The filter values are not designed manually; they are learned automatically by back-propagation during training. Figure 3 shows how a vertical-edge filter responds strongly at the position where a bright region meets a dark region, which is the same principle used to detect the border of an opacity in the lung.

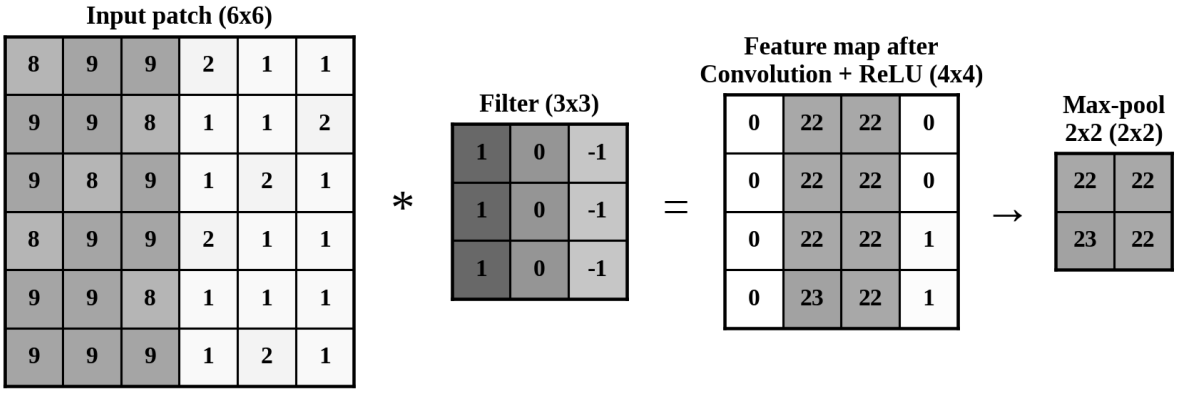


Figure 3: Convolution with a 3x3 filter, ReLU and 2x2 max pooling (worked example)

Pooling layers. A max-pooling layer (2x2, stride 2) keeps only the strongest response in every 2x2 region of a feature map, as seen in the last panel of Figure 3. This halves the width and height, so the computation and the number of parameters in later layers are reduced. It also makes the network tolerant to small shifts or tilts of the patient's position in the X-ray and

reduces over-fitting. In the proposed model, the spatial size shrinks from 224 x 224 to 14 x 14 while the depth grows from 1 to 256 channels, which means that the network moves from “where” information to “what” information.

Other components. ReLU introduces non-linearity and keeps the training fast. Global average pooling converts every feature map into one value, avoiding a very large dense layer. Dropout (0.5) randomly switches off neurons during training to prevent over-fitting. The softmax layer converts the final scores into probabilities of Normal and Pneumonia.

5. Algorithm

1. **Collect data:** Load the labelled chest X-ray images (Normal and Pneumonia).
2. **Pre-process:** Resize every image to 224 x 224, convert to grayscale and scale pixel values to the range 0 to 1.
3. **Augment:** Apply random rotation, zoom, horizontal shift and brightness change to the training set only; use class weights to handle the imbalance between Normal and Pneumonia images.
4. **Split:** Divide the data into training (about 70%), validation (about 15%) and test (about 15%) sets.
5. **Build the CNN:** Stack four blocks of Convolution + ReLU + MaxPooling, then global average pooling, a dense layer with dropout and a softmax output layer.
6. **Train:** Forward propagate each batch, compute the cross-entropy loss, back-propagate the error and update the filter weights using the Adam optimiser (learning rate 0.0001).
7. **Validate:** Monitor validation loss after every epoch; use early stopping and learning-rate reduction to avoid over-fitting.
8. **Test:** Evaluate on unseen images using accuracy, precision, recall (sensitivity), F1-score, confusion matrix and ROC-AUC. Recall is given priority because missing a pneumonia case is more harmful than a false alarm.
9. **Explain:** Generate Grad-CAM heat-maps to show the regions that influenced the prediction.
10. **Deploy:** Integrate the trained model into a clinic or tele-health application, which returns the predicted class with a confidence score to the radiologist for final confirmation.

6. Expected Outcome

- **Faster diagnosis:** The prediction is produced within seconds, so treatment can start early.
- **Wider access to care:** Rural and remote health centres without a radiologist can screen patients and refer only the critical cases.
- **Reduced workload and errors:** Doctors get a consistent second opinion, which lowers the chance of missed cases.
- **Lower cost:** Automated screening reduces the cost and waiting time of reporting.
- **Reduced mortality:** Early detection of pneumonia, especially in children and the elderly, helps to reduce preventable deaths, in line with SDG Targets 3.2 and 3.8.
- **Trust through transparency:** Grad-CAM heat-maps allow doctors to verify the model's reasoning.

The model must be validated on data from different hospitals and populations before clinical use, and it should only act as a support tool for qualified doctors.

7. Conclusion

The proposed CNN-based system shows that deep learning can automatically learn the visual signs of pneumonia from chest X-rays. Convolutional layers and filters extract meaningful features from edges to lung opacities, pooling layers reduce the data size and add robustness, and the fully connected softmax layer produces the final decision. By enabling earlier, cheaper and more widely available diagnosis, the solution contributes directly to SDG 3 by improving health outcomes and reducing preventable deaths, and it supports sustainable development by strengthening health-care systems in under-served areas.

Question 2: SDG 2 – Zero Hunger

Question: Develop a CNN-based approach for identifying plant leaf diseases to support sustainable agriculture. Explain how feature extraction and classification are performed using CNN.

1. SDG Identification

Selected SDG: SDG 2 – Zero Hunger.

Objective: To end hunger, achieve food security and improved nutrition, and promote sustainable agriculture. Target 2.3 aims to double the agricultural productivity of small-scale food producers, and Target 2.4 aims to ensure sustainable food production systems and resilient agricultural practices. Protecting crops from disease is essential to meet both targets.

2. Problem Statement

Plant diseases such as blight, rust, leaf spot and mould are a major threat to food security. According to the Food and Agriculture Organization (FAO), pests and diseases can destroy up to 40% of global crop production, which directly affects farmers' income and food availability. The main challenges are:

- Farmers usually identify diseases by naked-eye inspection, which needs experience and is often inaccurate because many diseases show similar symptoms.
- Agricultural experts are few and are not available in every village, so the disease is detected late and spreads over the field.
- Without a correct diagnosis, farmers spray pesticides excessively or wrongly, which increases cost, damages soil and water, and harms human health and biodiversity.

An automatic, quick and accurate disease identification tool that works on a smartphone is therefore required.

3. Proposed Solution

The proposed solution is a CNN-based image classification system. A farmer takes a photograph of an affected leaf with a smartphone; the CNN analyses the image and returns the crop disease name (or “healthy”) along with a confidence score and recommended treatment. The CNN is trained on a labelled leaf image dataset such as PlantVillage, which contains tens of thousands of images of healthy and diseased leaves of many crops, supplemented with real field images to improve robustness. For use on low-cost phones, the trained model can be converted to a lightweight format (for example TensorFlow Lite) and, where required, fine-tuned from a pre-trained network such as MobileNetV2 (transfer learning). The overall system is shown in Figure 4.

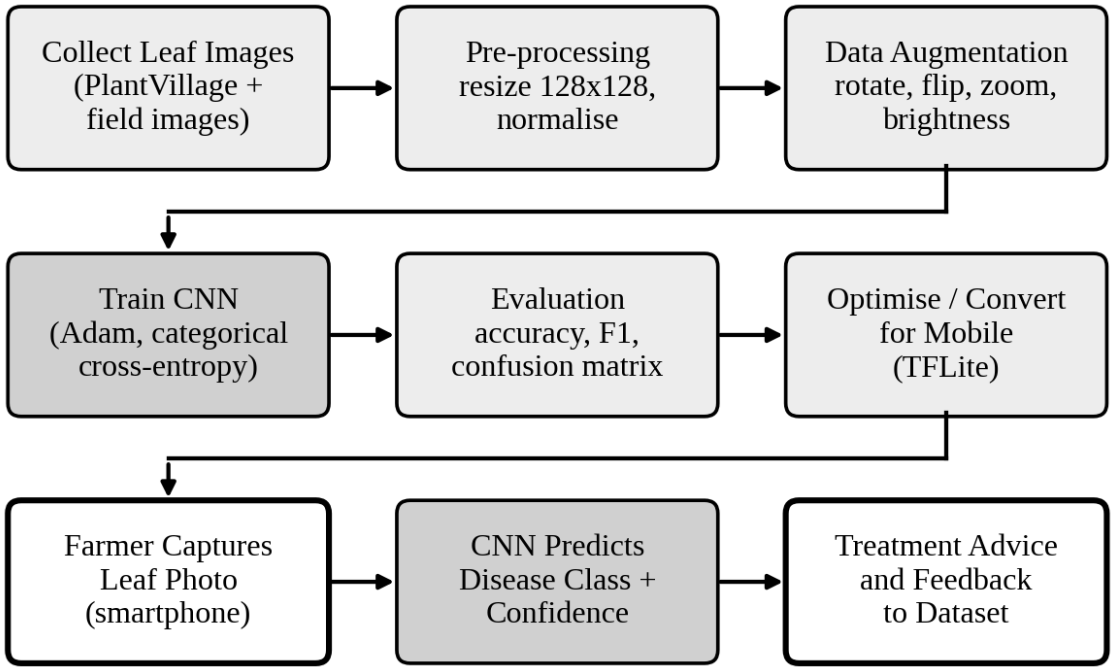


Figure 4: Workflow of the CNN-based plant leaf disease identification system

4. CNN Architecture

The network has two parts: (i) a **feature-extraction part** made of four convolution + max-pooling blocks, and (ii) a **classification part** made of a flatten layer, a fully connected layer with dropout and a softmax output layer. The architecture is shown in Figure 5 and Table 2.

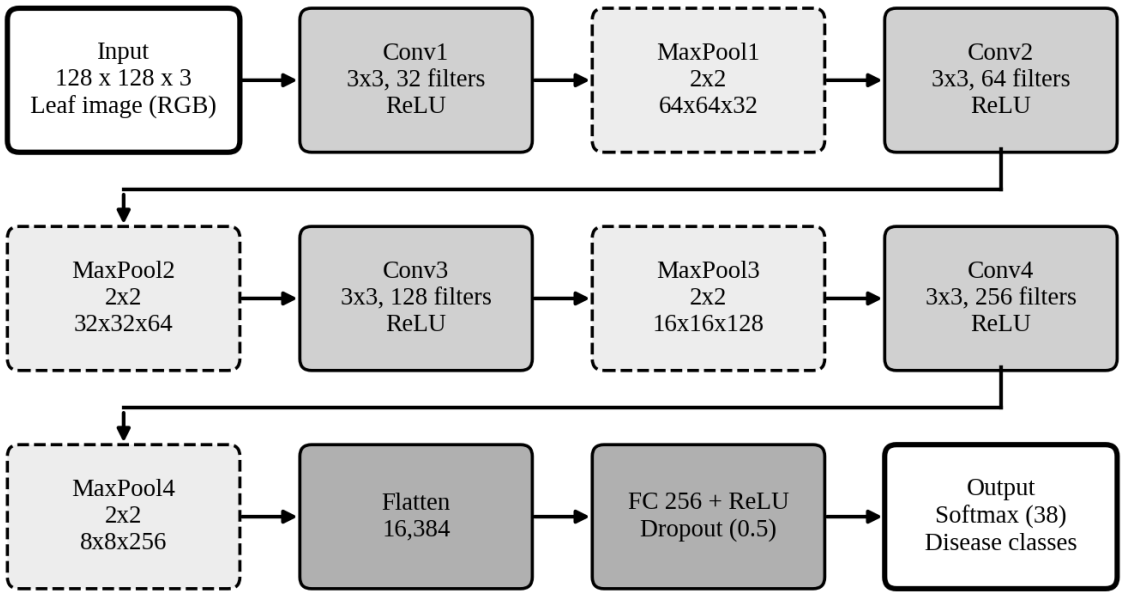


Figure 5: Proposed CNN architecture for plant leaf disease identification

Table 2: Layer-wise configuration of the proposed model

Layer	Configuration	Output Shape	Parameters
Input	RGB leaf image	128 x 128 x 3	0
Conv1 + ReLU	32 filters, 3x3, same padding	128 x 128 x 32	896
MaxPool1	2x2, stride 2	64 x 64 x 32	0
Conv2 + ReLU	64 filters, 3x3	64 x 64 x 64	18,496
MaxPool2	2x2, stride 2	32 x 32 x 64	0
Conv3 + ReLU	128 filters, 3x3	32 x 32 x 128	73,856
MaxPool3	2x2, stride 2	16 x 16 x 128	0
Conv4 + ReLU	256 filters, 3x3	16 x 16 x 256	295,168
MaxPool4	2x2, stride 2	8 x 8 x 256	0
Flatten	Converts maps to a vector	16,384	0
Dense + Dropout	256 neurons, ReLU, dropout 0.5	256	4,194,560
Output	Softmax, 38 classes	38	9,766

Layer	Configuration	Output Shape	Parameters
Total			**4,592,742**

Feature Extraction using CNN

Feature extraction is carried out by the convolution and pooling layers. In each convolutional layer, many 3x3 filters slide over the image and produce feature maps; ReLU then removes negative values. The CNN builds features in a hierarchy, as illustrated in Figure 6:

- **Early layers** (Conv1, Conv2) detect low-level features such as edges, leaf boundaries, veins and colour changes (for example green to yellow or brown).
- **Middle layers** (Conv3) combine them into textures and local patterns such as spots, rings, powdery coatings and holes.
- **Deep layers** (Conv4) detect high-level, disease-specific shapes such as the size, spread and arrangement of lesions that separate one disease from another.

After each convolution, max pooling keeps the strongest response in every 2x2 region, which reduces the image size from 128 x 128 to 8 x 8 and makes the features independent of the exact position, angle or size of the lesion on the leaf. Parameter sharing means that the same filter detects a spot wherever it appears.

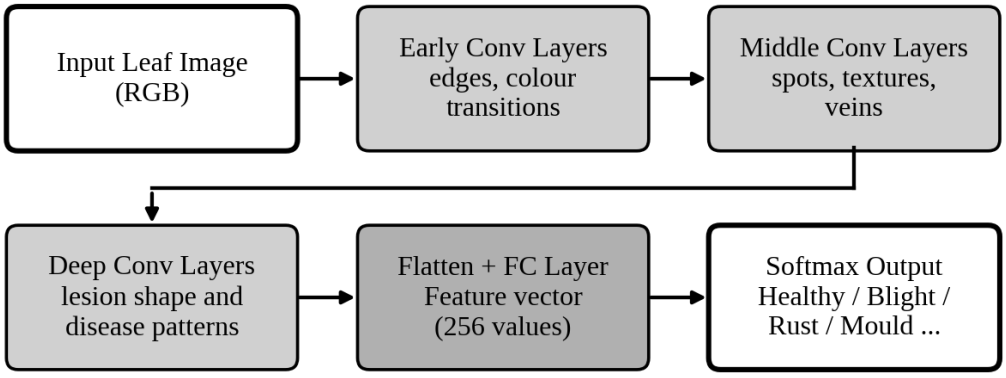


Figure 6: Hierarchical feature extraction and classification in the CNN

Classification using CNN

The 8 x 8 x 256 feature maps are flattened into a vector of 16,384 values, which is passed to a fully connected layer of 256 neurons. This layer combines all the extracted features to decide which disease pattern is present. Dropout (0.5) prevents over-fitting. Finally, the softmax layer with 38 neurons converts the scores into probabilities that sum to 1; the class with the highest probability is the predicted disease. The network is trained by minimising the categorical cross-entropy loss with the Adam optimiser, and the filter and neuron weights are updated by back-propagation.

5. Algorithm

1. **Collect data:** Gather labelled images of healthy and diseased leaves (for example the PlantVillage dataset) and add field images captured under natural light.
2. **Pre-process:** Resize each image to 128 x 128, normalise the pixel values to 0–1 and remove mislabelled or blurred images.
3. **Augment:** Apply rotation, flipping, zooming, shifting and brightness changes to the training data so that the model works under different field conditions.
4. **Split:** Divide the data into training (70%), validation (15%) and test (15%) sets.
5. **Build the CNN:** Create four Convolution + ReLU + MaxPooling blocks followed by Flatten, Dense (256), Dropout (0.5) and a Softmax output layer.
6. **Feature extraction:** Pass each image through the convolution and pooling layers to obtain high-level feature maps.
7. **Classification:** Feed the flattened features to the dense and softmax layers to obtain class probabilities.
8. **Train:** Compute the categorical cross-entropy loss, back-propagate the error and update the weights using Adam; use early stopping on the validation loss.
9. **Evaluate:** Measure accuracy, precision, recall, F1-score and the confusion matrix on the test set.
10. **Deploy:** Convert the model for mobile use. The farmer uploads a leaf photo, receives the disease name, confidence and treatment advice, and the confirmed new images are added to the dataset for further improvement.

6. Expected Outcome

- **Early disease detection:** Farmers can identify the disease at the first symptoms and stop it from spreading, which protects the harvest.
- **Higher yield and food security:** Reducing crop losses increases the food available and farmers' income, supporting SDG Targets 2.3 and 2.4.
- **Reduced pesticide use:** Correct diagnosis allows targeted treatment instead of blind spraying, which lowers cost and protects soil, water and biodiversity.
- **Expert knowledge for everyone:** A smartphone application gives small and remote farmers access to expert-level diagnosis at almost no cost.
- **Scalability:** The same approach can be extended to new crops and diseases by adding new images and retraining.

Since accuracy on laboratory images can be higher than on real fields, the system should be tested and retrained with field data from different regions, seasons and lighting conditions.

7. Conclusion

The proposed CNN-based approach automatically extracts features from leaf images, from edges and colours to lesion patterns, and classifies them into healthy or specific disease categories using fully connected and softmax layers. It provides farmers with fast, accurate and affordable disease identification, reduces crop losses and unnecessary pesticide use, and thereby contributes to SDG 2 by improving food security and promoting sustainable agriculture.